# Data Visualisation and Data-driven Decision Making Project

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Project Webpage | GitHub Repository

## 1 Introduction

We present a webpage containing four main visualisations of historic stock market data (Defeat Beta, 2024) from January 2019 to December 2025. These visualisations are realized entirely with the python matplotlib module, and are ordered in terms of data granularity and are meant to represent different views of the stock market, from subjective personal feelings to objective hard realities. This aims to to bridge the mental gap between a trader looking at their personal portfolio performance and the broader trends of the market, allowing for easier rationalization of trading decisions. The first visualisation is an interactive exercise, showing the disparity between short-term and long-term single-stock performance through the use of line charts. The second visualisation is a heatmap of aggregate industry monthly performance meant to express the waves of gains and losses. The third is a bar chart showcasing quarterly sector performance, revealing trends of the broader market, while allowing for comparison between medium- and long-term returns. Lastly, an animated spiral line chart shows the overarching reality of aggregate market performance.

## 2 Data Pre-processing

Data is first adjusted for inflation to the 2025 US Dollar. The data is then filtered based on the starting closing price of each stock and the average trading volume. Stocks above the 20th percentile of either their industry, sector, or global group, depending on the plot, are retained. This prevents penny stocks from skewing the average price and change in percentage, especially as their market importance is highly negligible and only relevant for a small group of people and businesses.

## 3 Chart Decisions and Justification

When actively engaging in trading, it is hard to rationalize the relatively short-term losses while comparing them with the overall portfolio performance: a sudden 5% decrease from the previous week's values can eclipse the overall 15% return for the current year, as well as casting doubts over one's trading choices, not realizing the possibility of a broader market slowdown. Therefore, we decide to visualise this juxtaposition of subjective day-to-day feelings and objective, longer-term, realities to implore the reader to stay level-headed when faced with short-term challenges, whether they arise trading stocks or during everyday life. The main colors [Figure 1] used in plots were chosen to be colorblind friendly.

### 3.1 Short-term Long-term Disparity Exercise

With this exercise, we use multiple single-stock line charts to try to model the day-to-day experience of trading stocks; looking at bi-weekly return lines rising and falling, attempting to predict which stocks will be beneficial over the longer term, embodied by the 2019-2025 line charts, while trying to remain optimistic that prior choices will result in profit, and not loss, but also grounded when short-term spikes occur.

### 3.2 Daily Industry Performance Heatmap

Showcases the waves of gains and losses across industries, showing the slightly broader and longer term context that is effecting individual stocks. Thus, the plot allows for more rational, and educated, reasoning regarding single-stock performance. To focus on the bigger picture, not the specific details, the industry labels were omitted.

### 3.3 Quarterly Sector Performance Bars

Shows broad market trends per sector, both quarter-to-quarter and all-time (2019-2025). This level of granularity now shows real market trends, with negative quarters likely being indicators of systematic regression in price, rather than a temporary negative moment for specific industries or stocks, adding more context to the longer-term performance. The viewer is encouraged to draw conclusions about the overall profitability of a given sector based on quarterly data, and subsequently compare their intuition with the information in the all-time chart.

### 3.4 Overall Animated Spiral of Performance

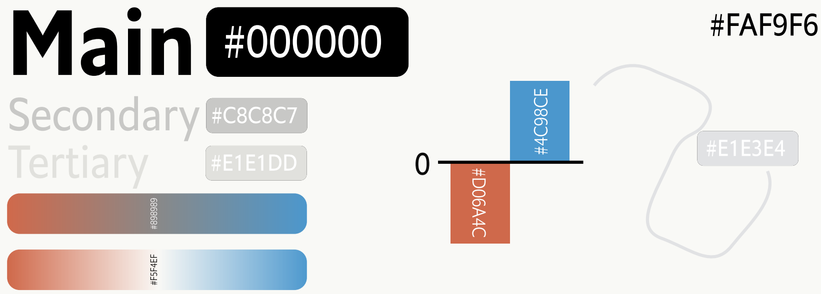
Made as an alternative to a stacked line chart, the spiral facilitates the comparison of daily aggregate stock trends and performance across years using a continuous line, rather than multiple disconnected lines, allowing for animation. It shows the highs and lows of trading both inside individual years, and over multiple years. This provides full long-term context about the gains or losses of sectors and industries in any period of time, while also serving as a point of comparison between the performance of one's own portfolio and the average market performance.

## References


Defeat Beta (2024). Yahoo finance data dataset. https://huggingface.co/datasets/defeatbeta/yahoo-finance-data. Accessed: 15-04-2026, Licensed under ODC-BY.


## A Color Pallete

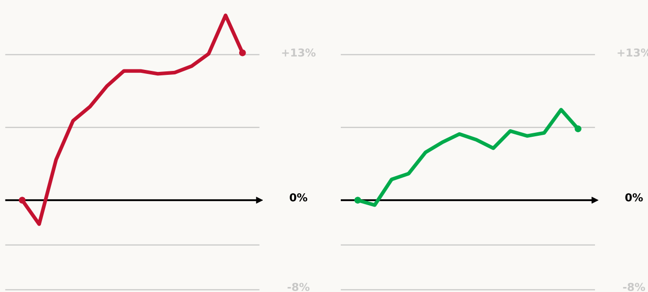
Figure 1: Color Pallete



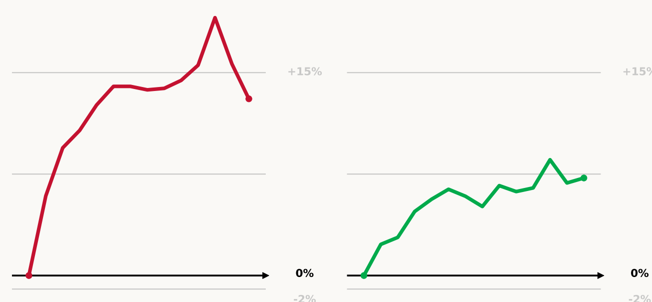

## B Plot samples

All interactive elements, animated plots, and full versions of plots can be found on the Project Webpage, full versions of plots can also be found in the GitHub Repository

Figure 2: Short-term Long-term Disparity Exercise




(a) Sample A




(b) Sample B


(c) Full stock sample

A button can be pressed to pick a random span of two weeks [Figure 2a,Figure 2b], the user then picks a perceived order of performance of the stocks at which point the full history is unlocked [Figure 2c]

Figure 3: Daily Industry Performance Heatmap

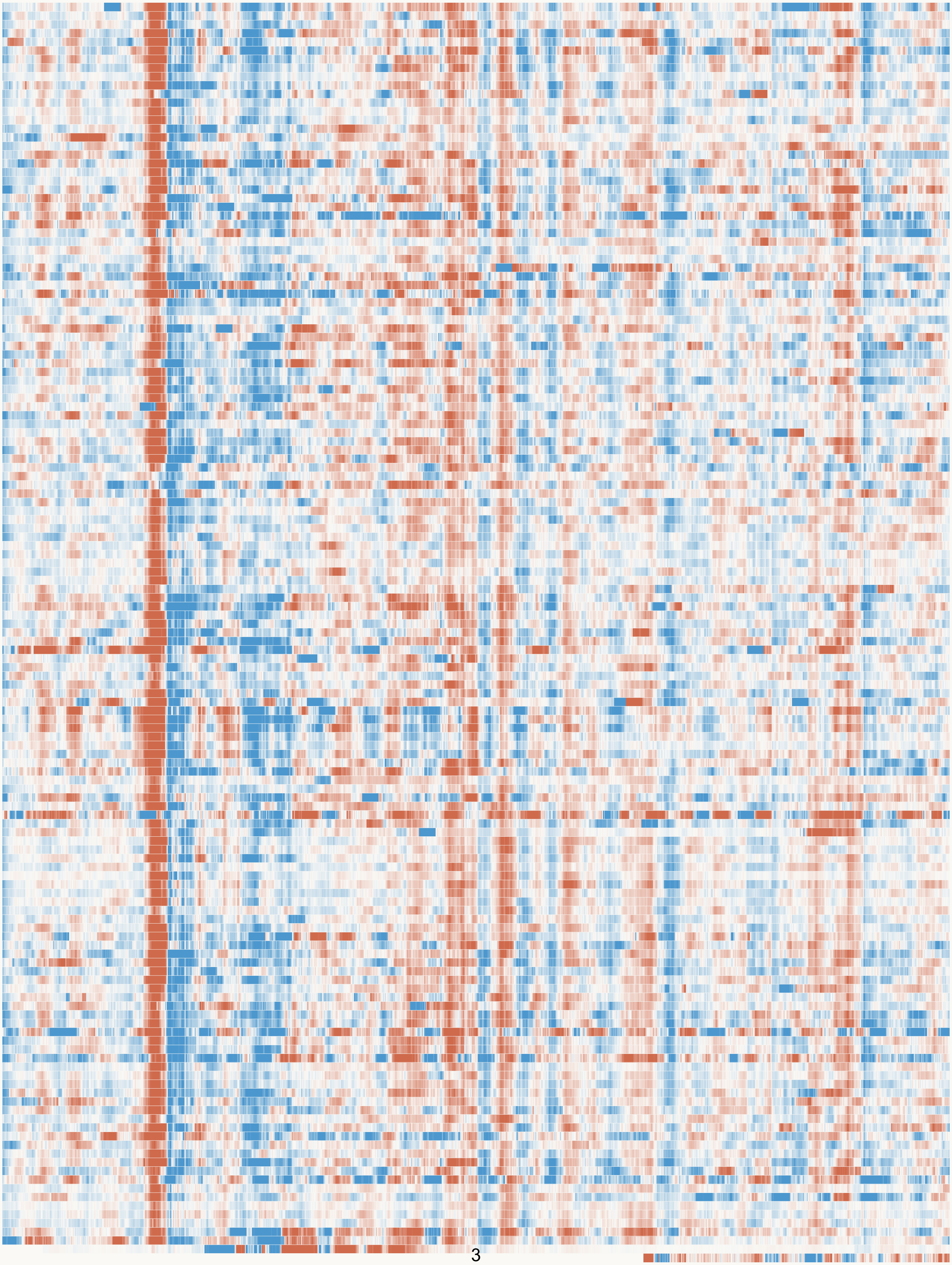
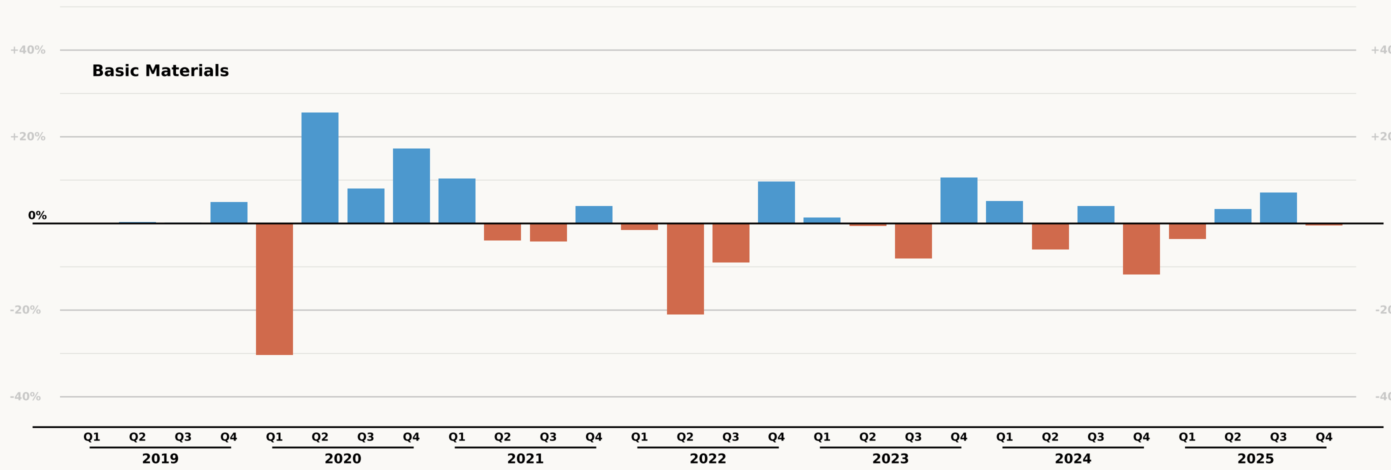
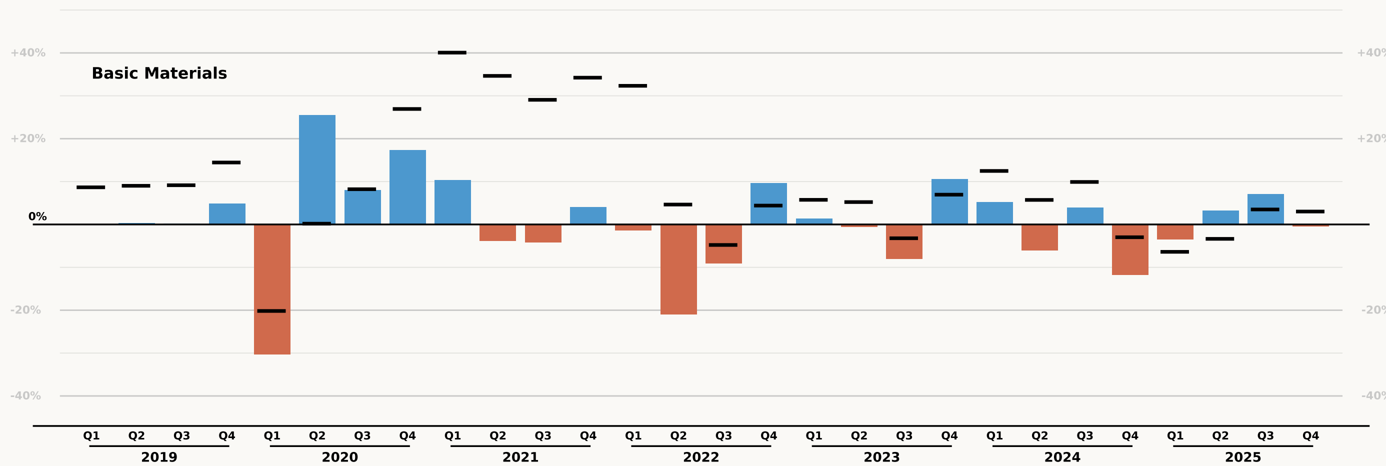


Figure 4: Quarterly Sector Performance Bars

(a) Without 2019 ticks.

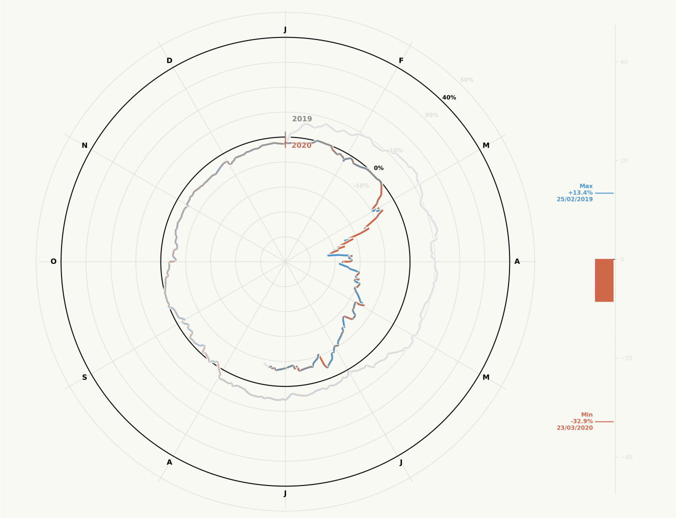



(b) With 2019 ticks.

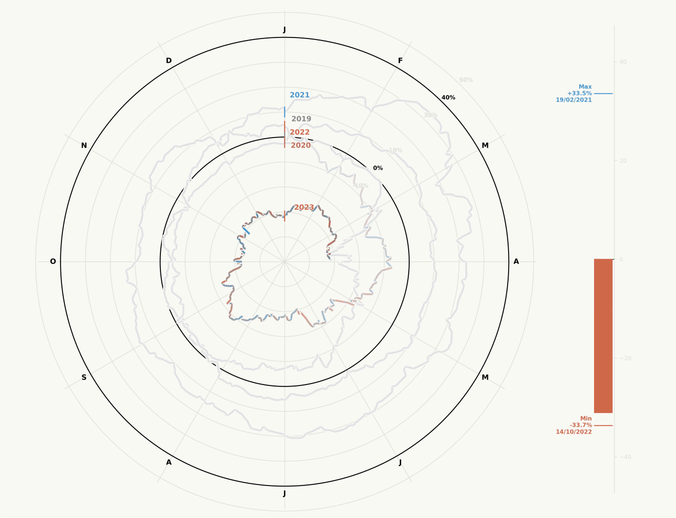



The interactive element allows for triggering of the quarterly comparison to 2019.

Figure 5: Overall Animated Spiral of Performance




(a) Sample 1




(b) Sample 2

The animation is the line being drawn as time passes as well as a bar indicating current return since beginning and maximum gain and loss. The color of the line is based on the daily gains and losses and fades into gray after a certain number of days.